

permanently marked after air balancing is completed so that these can be restored to their correct position if disturbed at any time. Complete air balance report shall be submitted for scrutiny and approval, and four copies of the approved balance report shall be provided with completion documents.

10. INSULATION

10.1 SCOPE

The scope of this section comprises the supply and application of insulation conforming to these specifications.

10.2 MATERIAL

- Insulation material shall be Closed Cell Elastomeric Nitrile Rubber with UV resistant metal finishing on one side.
- Thermal conductivity of insulation material shall not exceed 0.035 W/(m.K) at mean temperature of 0°C as per EN 12667.
- Moisture Diffusion Resistance Factor or 'μ' value of laminated insulation material shall be minimum ≥ 7000 as per EN 12086 for the base foam.
- Water Absorption by Volume of insulation material shall be < 0.2% as per ASTM C 1763 / ASTM C 209 for the base foam.
- The base insulation material shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame and also passes Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990.
- Density of laminated insulation material shall be between 50 to 70 Kg/m³.
- Density of base foam insulation material shall be between 40 to 55 Kg/m³.
- The Tensile strength of the base foam with covering >1.5 MPa (32mm) in MD & >0.4 MPa (32mm) in CD
- The Elongation of the base foam with covering >40% (32mm) in MD & >38% (32mm) in CD
- The Tear strength of the base foam with covering >38 KN/m (32mm) in MD & 27 KN/m (32mm) in CD
- The insulation material shall be dust and fibre free.
- The insulation material shall be formaldehyde free.
- The insulation material shall be CFC & HCFC free.
- The insulation material shall withstand maximum surface temperature of +85°C and minimum surface temperature of 0°C as per EN 14706.
- The base insulation material shall have ODP (Ozone Depletion Potential) and GWP (Global Warming Potential) of Zero.
- Covering material shall be of multiple layered laminate of polymeric material reinforced with scrim with a special UV protection.
- Covering material shall be non metallic.
- Covering material shall provide mechanical resistance with an excellent aesthetic look.
- Covering material shall be mat finish.
- Covering material shall be of 380 ± 20 GSM as per ASTM D 646.
- Covering material shall be of 380 – 400 microns as per ASTM E 252
- Covering material shall have the tensile strength of >5 MPa in MD & CD as per ASTM D 638

complies
(material density
will be 40-60
kg/m³)

Complies,
(GSM will
be 512,
thickness
will be 350
microns)

- Covering material shall have good puncture and tear resistance.
- Covering material shall have good UV resistance as per ISO 4892-2 Method A tested for 1000 Hrs.
- Covering material shall withstand maximum surface temperature of +75 °C and minimum surface temperature of -25 °C as per EN 14706.
- Covering material shall be installed with 30mm – 50mm overlaps on all horizontal and longitudinal joints. Joints along the material length shall be installed facing downward. The joints shall be secured with 50 mm – 75 mm tapes. Manufacturer recommended adhesive can be used for flat sheets, or where restricted access prevents use of rivets. All joints and overlaps must be finished with manufacturer recommended PSA Tape.
- For fittings of an unusual shape, or large size, the covering material should be cut and installed as per traditional metal cladding.

Insulation thickness values will be selected as per following selection table.

Service	Pipe diameter	Insulation Thickness (mm)
Chilled water system pipe	15mm – 80mm	32
	100mm – 500mm	51
	600mm – 1200mm	75
Medium water system pipe	25 mm – 350mm	32
Process water system pipe	25 mm – 350 mm	32

noted

Ducting insulation thickness shall be as per table below.

Ducting position	Insulation Thickness
SA duct	32mm
Return air system	25mm

noted

10.3 PUMP INSULATION

Chilled water pump shall be insulated to the same thickness as the pipe to which they are connected and application shall be same as above. Care shall be taken to apply insulation in a manner as to allow the dismantling of pumps without damaging the insulation.

10.4 SHELL INSULATION

The chiller shells shall be factory insulated in accordance with the manufacturer's standards.

10.5 CHILLED WATER EXPANSION TANK INSULATION

Cold water tank, and chilled water expansion tank shall be insulated as per manufacturer's standard.

noted

10.6 ACOUSTIC LINING OF MECHANICAL ROOMS

Two walls and ceiling of air conditioning plant room and air handling unit rooms may be provided with acoustic lining. Material shall be processed from Nitrile rubber foam.

complies
(density will be
180-220 kg/m³)

The material should be fiber free. The density of the same shall be 140-180 Kg/m³. The material should have thermal conductivity not exceeding 0.05 W/Mk. The maximum surface temperature of material shall withstand is 1050C. and minimum temp shall be - 200C. Thickness shall be as specified. The material should confirm to class 1 rating for surface spread of flame as per BS 476 Part 7. Thickness, if not specified, shall be considered as 20mm.

Surface shall be cleaned and adhesive recommended by the manufacturer should be applied on the walls. The foam sheets should be cut to required size and a layer of adhesive should also be applied to it. When it is tack dry it is stuck to the walls / ceiling. Acoustic lining of walls shall be terminated approximately 15 cm above the finished floor to prevent damage to insulation due to accidental water-logging in plant/AHU rooms.

10.7 SOUND ATTENUATORS

- a) Attenuators shall be installed in ducts in accordance with requirements of drawings and as included in Schedule of Quantities.
- b) Attenuators shall be of steel construction with casings out of minimum
- c) 22 G galvanized steel. Acoustic fill shall be inert, non-hygroscopic, vermin proof, fiber glass of required density adequately protected against corrosion and covered with 26 gage perforated aluminum sheet. Attenuators shall be supplied complete with flanges.
- d) The pressure drop values of the silencers shall be indicated for each duty.
- e) Manufacturers shall submit a test certificate for acoustic and aerodynamic performance of the attenuators. Attenuators shall be tested in accordance with ACMA test methods/BS 4718 and insertion loss and self-generated noise for each octave band and pressure drop shall be stated in the schedule.

10.8 MEASUREMENT OF INSULATION

Unless otherwise specified measurement for duct and pipe insulation for the project shall be on the basis of center line measurements described herewith

- a) Pipe Insulation shall be measured in units of length along the center line of the installed pipe, strictly on the same basis as the piping measurements described earlier. The linear measurements shall be taken before the application of the insulation. It may be noted that for piping measurement, all valves, orifice plates and strainers are not separately measurable by their number and size. It is to be clearly understood that for the insulation measurements, all these accessories including cladding, valves, orifice plates and strainers shall be considered strictly by linear measurements along the center line of pipes and no special rate shall be applicable for insulation of any accessories, fixtures or fittings whatsoever.
- b) Duct Insulation and Acoustic Lining shall be measured on the basis of surface area along the center line of insulation thickness. Thus the surface area of externally thermally insulated or acoustically lined be based on the perimeter comprising center